Data Codebook

Final municipality-year dataset

The variables used in our final dataset are described below. Our final dataset was created based on two Statistics Finland resources: Municipal Key Figures and Establishments of enterprises. The data processing steps include data cleaning in MATLAB, retaining only municipality observations, merging the datasets by municipality code and year, omitting missing values, performing correlation analysis and detecting outliers, and saving our final data to work further with the tasks of the project.

The final dataset is a municipality-year dataset. Each row is one municipality in one year. The data cover 2018-2023, because 2024 was removed after checking missing values in workplace-structure variables.

Item	Details
Final files	Municipal_key_figures_final.xlsx and Municipal_key_figures_final.csv
Rows and columns	1847 rows and 22 columns
Observation unit	Municipality-year observation
Municipalities	308 municipalities
Years	2018-2023
Missing values	0 missing values in the final dataset
Join key	RegionCode and Year in the raw data; MunicipalityCode and Year in the final data

Short Notes on Cleaning

- Only municipality codes starting with KU were retained in the final dataset. All other regional and administrative aggregates were omitted.

- Data from Municipal Key Figures were initially in long format. To transform the data, the unstack command in MATLAB was used.

- The two initial datasets were merged by municipality code and year.

- The year 2024 was eliminated since some selected workplace structure indicators were missing for all municipalities.

- Highly correlated raw scale variables were omitted from the final set of predictors: EmployedPeople, Personnel, Turnover, and WorkingAgeShare.

- While outlier detection was performed, the outliers were not automatically omitted, since most of those extremely large or small values reflected real differences between Finnish municipalities.

How Identifiers Can Be Used

MunicipalityCode, Municipality, and Year are identifiers, so they are not treated as normal predictors in the supervised model. However, they are still useful in the analysis. MunicipalityCode and Municipality help us label results, compare municipalities, and interpret clusters. Year helps us summarize the data by time, compare changes from 2018 to 2023, and check whether the patterns are similar across years.

Variable Explanations

The table below explains each variable in the final dataset. Identifier variables are kept for labelling and interpretation. Predictors are used as explanatory variables. UnemploymentRate is the target variable for supervised modelling.

Variable	Role	Data Source	Description
MunicipalityCode	Identifier	Statistics Finland	Code for a specific municipality. It was used in identifying the unique municipalities and merging of the two datasets. In analysis, MunicipalityCode will help in grouping observations by municipalities, checking the presence of every municipality and linking results to the correct municipality.
Municipality	Identifier	Statistics Finland	Name of the municipality. This information was retained to make the results easier to understand and interpret. In analysis, Municipality can be used for labeling purposes in tables, figures, mapping and cluster identification to differentiate which municipalities belong to which clusters.
Year	Identifier	Statistics Finland	Year when the observation was made. The final dataset included municipality and year combinations ranging from 2018-2023. In analysis, Year can be used for comparison and grouping of observations in annual periods.
UrbanShare	Predictor	Municipal Key Figures	Degree of urbanisation in percent. This describes how urban the municipality is.
DependencyRatio	Predictor	Municipal Key Figures	Economic dependency ratio. This gives an idea of demographic and economic pressure in the municipality.
WorkInOwnArea	Predictor	Municipal Key Figures	Share of employed people who work in their own municipality of residence. This helps describe local work opportunities and commuting.
EmploymentRate	Predictor	Municipal Key Figures	Employment rate of people aged 20 to 64. This is one of the main labour-market indicators.
ForeignCitizens	Predictor	Municipal Key Figures	Foreign citizens as a percentage of the population. This describes part of the population structure.
RentalHousingShare	Predictor	Municipal Key Figures	Share of household-dwelling units living in rental dwellings. This also demonstrates the urban structure and housing situation.
NetMigration	Predictor	Municipal Key Figures	Intermunicipal migration gain or loss, expressed in persons. This indicates if the municipality experiences gains or losses due to intermunicipal migration.
OlderShare64	Predictor	Municipal Key Figures	Share of people aged 64 or over. We kept this as an ageing indicator.
YouthShare15	Predictor	Municipal Key Figures	Share of people aged under 15. This describes the younger population in the municipality.
TertiaryShare	Predictor	Municipal Key Figures	Share of people aged 15 or over with tertiary level qualifications. This represents higher education level.
PopulationChange	Predictor	Municipal Key Figures	Population change from the previous year, in percent. This shows whether the municipality is growing or declining.
PrimaryJobs	Predictor	Municipal Key Figures	Share of jobs in primary production. This describes the importance of primary-sector work.
SecondaryJobs	Predictor	Municipal Key Figures	Share of jobs in secondary production. This describes the industrial and manufacturing job structure.
ServiceJobs	Predictor	Municipal Key Figures	Share of jobs in services. This describes how service-oriented the municipality job structure is.
UpperSecondaryShare	Predictor	Municipal Key Figures	Share of people aged 15 or over with at least upper secondary qualifications. This is used as a general education indicator.
WorkplaceSelfSufficiency	Predictor	Municipal Key Figures	Workplace self-sufficiency. This shows how many jobs are located in the municipality compared with the employed labour force living there.

Variable	Role	Data Source	Description
Establishments	Predictor	Establishments of enterprises	Number of enterprise establishments. We kept this as a simple business activity indicator.
TurnoverPerPerson	Predictor	Establishments of enterprises	Turnover of establishments per person, measured in EUR 1,000. We kept this instead of raw turnover because it is easier to compare across municipalities.
UnemploymentRate	Target	Municipal Key Figures	Proportion of unemployment among the labour force. This is the target variable in our supervised modelling and cannot be used as a predictor.

An important note for modelling: UnemploymentRate should not be used as a predictor when building a model.